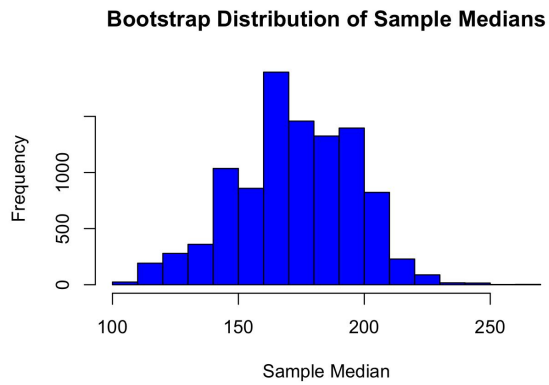
STAT 5703 **HW 2** Spring 2026

Solution

1. We are interested in estimating the median time it takes people to fill out a tax form. The data are in the file `irs.txt`.

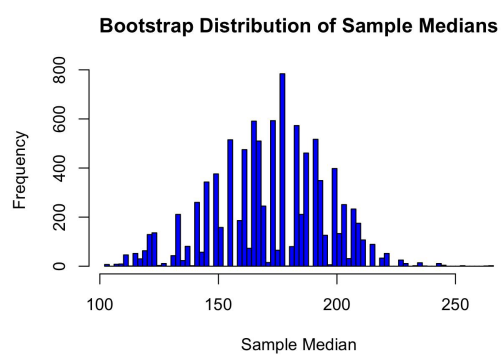
   a) Obtain a bootstrap distribution of the sample median and make a histogram of it.

```
times <- read.table('irs.txt', header = TRUE)$Time

# Bootstrap medians
bootstrap_median <- function(data, n_bootstrap) {
  bootstrap_medians <- replicate(n_bootstrap, median(sample(data, length(data), replace = T)))
  return(bootstrap_medians)
}

# Perform
n_bootstrap_samples <- 10000
sample_medians <- bootstrap_median(times, n_bootstrap_samples)

# Create the histogram
hist(sample_medians, main = "Bootstrap Distribution of Sample Medians",
     xlab = "Sample Median", col = "blue")
```



   b) What seems odd about it? Why did that happen?
   Hint: Change number of bins on your plot, try 50 and try less.

```
hist(sample_medians, breaks = 65, main = "Bootstrap Distribution of Sample Medians",
xlab = "Sample Median", col = "blue")
```



When looking at the histogram, you might notice that the distribution of medians is not smooth. This usually happens when there are a lot of repeated values in the data, or the sample size is not large enough. In addition, the median is not very sensitive to small changes in the sample.

   c) Choose and report any 95% confidence interval for the population median.

```
(conf_int <- quantile(sample_medians, probs = c(0.025, 0.975)))
2.5% 97.5%
```

121 212

2. Let $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$ be an iid sample. We want to test H_0: $\lambda = \lambda_0$ vs. H_a: $\lambda = \lambda_1$, with $\lambda_1 > \lambda_0$.

   a) Derive the form of the rejection region at level α with Neyman-Pearson approach.

The Exponential pdf is $f(x;\lambda) = \lambda e^{-\lambda x}$

$$\Rightarrow L(\lambda) = \prod_{i=1}^{n}\left(\lambda e^{-\lambda x_i}\right) = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$\Rightarrow \frac{L(\lambda_0)}{L(\lambda_1)} = \frac{\lambda_0^n e^{-\lambda_0 \sum_{i=1}^{n} x_i}}{\lambda_1^n e^{-\lambda_1 \sum_{i=1}^{n} x_i}} = \left(\frac{\lambda_0}{\lambda_1}\right)^n e^{(\lambda_1-\lambda_0)\sum_{i=1}^{n} x_i} \le k$$

$$\Leftrightarrow e^{(\lambda_1-\lambda_0)\sum_{i=1}^{n} x_i} \le k_1 \Leftrightarrow (\lambda_1-\lambda_0)\sum_{i=1}^{n} x_i \le k_2$$

$$\Leftrightarrow \sum_{i=1}^{n} x_i \le c$$

Note that in the last step we divided both sides of the inequality with $\lambda_1 - \lambda_0 > 0$

   b) Assume $n = 4$, $\lambda_0 = 1$, $\lambda_1 = 4$, and $\alpha = 0.02$. Find the critical value and Type II error. *Hint:* Recall that sum of independent exponentially distributed variables with the same parameter λ has a Gamma distribution.

We need to find $P\left(\sum_{i=1}^{n} X_i \le c \middle| \lambda = 1\right)$, where $X_1, \ldots, X_n \sim \text{Exp}(1)$ be an iid sample. We can use the fact that if $X_1, \ldots, X_n \sim \text{Exp}(1)$ is an iid sample, then $\sum_{i=1}^{n} X_i \sim \Gamma(n, 1)$.

That is, to find c we need the lower 0.02 quantile of the $\Gamma(4, 1)$ distribution. Using R, the solution is

$$c = \text{qgamma}(0.02, 4, 1) = 1.016$$

For the Type II error, we need $P\left(\sum_{i=1}^{n} X_i > c \middle| \lambda = 4\right) = P(\Gamma(4, 4) > 1.016)$

In R, the solution is:

$$1 - \text{pgamma}(1.016, 4, 4) = 0.421$$

   c) Let the observed sample be 1.2, 3, 1.5, 0.3 and we wish to test H_0: $\lambda = 1$ vs. H_a: $\lambda = 4$, at $\alpha = 0.02$ level of significance. Compute the p-value and state your conclusion.

$$\sum_{i=1}^{n} x_i = 6$$

The p-value of a hypothesis test is the probability of obtaining a value of the test statistic which is more extreme than the one observed in the current sample, assuming H_0 is true. Here,

$$\begin{aligned}
\text{p-value} &= P(\sum_{i=1}^{n} X_i \leq 6 | H_0) \\
&= P(\Gamma(4,1) \leq 6) \\
&= F_{\Gamma(4,1)}(6) \\
&= 0.848
\end{aligned}$$

Since the p-value$> \alpha = 0.02$, we fail to reject H_0 at a level of significance of 0.02

3. Let $X_1, X_2, \ldots, X_n \sim N(\mu, \sigma)$ be and iid sample, where the mean μ and the standard deviation σ are *both* unknown.

a) Use $\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$ as a pivot and derive a formula for (1 - α)100% CI for σ^2.

Let xL and xU be the lower and upper α/2 quantiles of the χ^2_{n-1} distribution. Then we have that

$$P\left(x_L \le \frac{(n-1)s^2}{\sigma^2} \le x_U\right) = 1-\alpha$$

$$\Leftrightarrow P\left(\frac{1}{x_U} \le \frac{\sigma^2}{(n-1)s^2} \le \frac{1}{x_L}\right) = 1-\alpha$$

$$\Leftrightarrow P\left(\frac{(n-1)s^2}{x_U} \le \sigma^2 \le \frac{(n-1)s^2}{x_L}\right) = 1-\alpha$$

Therefore, $\left[\frac{(n-1)s^2}{x_U}, \frac{(n-1)s^2}{x_L}\right]$ is a (1 - α)100% CI for σ^2.

b) Suppose $n = 63$, $\bar{x} = 16.7, s = 3.3$. Compute a 95% CI for μ.

Since σ is unknown, the (1 - α)100% CI for μ is

$$\bar{x} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}} = 16.7 \pm 1.99897 \times \frac{3.3}{\sqrt{63}}$$
$$= (15.8689, 17.531)$$

*Personally, I think it is reasonable to use normal distributions to approximate the t values because n is large.

c) Suppose $n = 63$, $\bar{x} = 16.7, s = 3.3$. Compute a 94% CI for σ^2.

From part a), we have that a (1 - α)100% CI for σ^2 is

$$\left[\frac{(n-1)s^2}{x_U}, \frac{(n-1)s^2}{x_L}\right] = \left[\frac{(63-1)(3.3)^2}{\chi^2_{0.03,62}}, \frac{(63-1)(3.3)^2}{\chi^2_{0.97,62}}\right]$$
$$= [7.98, 15.77]$$

4. (This continues from HW 1, Problem 4). Let $X_1, \ldots, X_n$ be an iid sample from a distribution with probability mass function $f(x;\alpha,\beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot x^{\alpha-1}(1-x)^{\beta-1}, x \in (0,1)$, where the unknown parameter is a vector $\boldsymbol{\theta} = (\alpha, \beta)$ with $\alpha > 0$, and $\beta > 0$. Suppose the observed sample is $(x_1, x_2, x_3, x_4, x_5) = (0.34, 0.47, 0.42, 0.23, 0.26)$.

a) Construct approximate 95% Wald CIs for α and β.

From HW1, the Fisher information matrix is:

$$I_n(\alpha, \beta) = n \begin{bmatrix} \psi_1(\alpha) - \psi_1(\alpha+\beta) & -\psi_1(\alpha+\beta) \\ -\psi_1(\alpha+\beta) & \psi_1(\beta) - \psi_1(\alpha+\beta) \end{bmatrix}$$

where $\psi_1(z) = \frac{d^2}{dz^2} \log \Gamma(z)$ is the `trigamma` function.

Also, we found that $\widehat{\theta} = (9.213088, 17.570113)$.
Plugging in the Fisher information matrix, we obtain that

$$\hat{I}_n = \begin{bmatrix} 0.383 & -0.19 \\ -0.19 & 0.1 \end{bmatrix} \Rightarrow \hat{I}_n^{-1} = \begin{bmatrix} 32.878 & 60.95 \\ 60.95 & 122.72 \end{bmatrix}$$

Therefore, the 95% Wald CI for α is

$$(9.213 \pm 1.96\sqrt{32.88}) = (-2.03, 20.45)$$

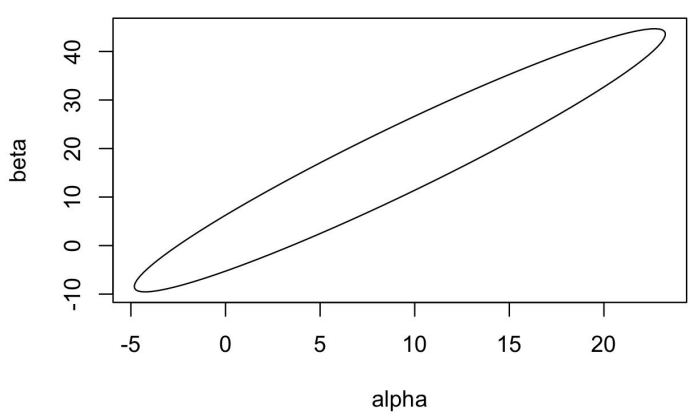
and the 95% Wald CI for β is

$$(17.57 \pm 1.96\sqrt{122.72}) = (-4.14, 39.28)$$

Note that the CI include negative values which is impossible, because the parameters must be positive. This is happening since we are applying an asymptotic method to a sample of size only 5.

b) Use software to draw a 95% confidence oval for the vector parameter $\boldsymbol{\theta}$.

```
library(ellipse)
plot(ellipse(solve(I), centre = ss$root), type = "l", xlab = "alpha", ylab =
"beta")
```



c) Suppose that you are told $\beta = 1$. Construct a 95% profile likelihood CI for α.

First derive the new likelihood:

$$L(\alpha) = \prod_{i=1}^{n} \left(\alpha x_i^{\alpha-1}\right) = \alpha^n \left(\prod_{i=1}^{n} x_i\right)^{\alpha-1}$$

So the log-likelihood is $l(\alpha) = n \log \alpha + (\alpha - 1) \sum_{i=1}^{n} \log x_i$.

Let $\frac{\partial l(\alpha)}{\partial \alpha} = 0 \Rightarrow \widehat{\alpha} = \frac{n}{\sum_{i=1}^{n} \log x_i}$.

Here $n = 5$, and $(x_1, x_2, x_3, x_4, x_5) = (0.34, 0.47, 0.42, 0.23, 0.26)$. Plug in these values and we will get $\widehat{\alpha} \approx 0.906$.

The 95% profile likelihood interval is the set of α satisfying $2\left(l\left(\widehat{\alpha}_0\right) - l(\alpha)\right) \leq \chi^2_{1,0.05} = 3.841$, or $l(\alpha) \geq l(\widehat{\alpha}_0) - 1.921$. Solving numerically yields the interval

$$\alpha \in (0.325,\ 1.947).$$